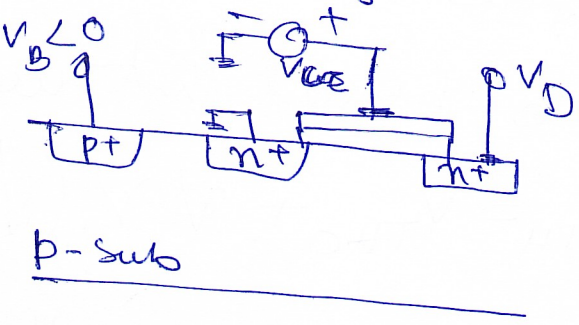
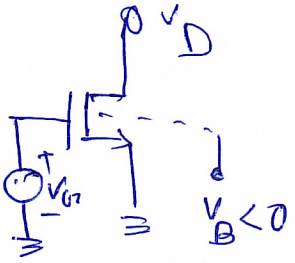
Second-order Effects:

Three second-order effects that are essential to our subsequent circuit analyses.

Body Effect:





On the analysis (earlier), we ~~~~ assumed that the bulk and source of the transistor were tied to ground. What happens if the bulk voltage of an NFET drops below the source voltage? Since the S and D junctions remain reverse-biased, we surmise that the device continues to operate properly but certain characteristics may change. To understand the effect, suppose $V_S = V_D = 0$, and V_{GS} is somewhat less than V_{TH} so that a depletion region is formed under the gate but no inversion layer exists. As V_B becomes more negative, more holes are attracted to the substrate connection, leaving a larger negative charge behind, ~~the~~ the depletion region becomes wider.

Now recall that the threshold voltage is a function of the total charge in the depletion region because the gate charge must mirror Q_d before an inversion layer is formed.

Thus, as V_B drops and Q_d increases, V_{TH} also increases. This is called the "body effect" or the backgate effect."

It can be proved that with body effect:

$$V_{TH} = V_{THO} + \gamma \left(\sqrt{|2\phi_F + V_{SB}|} - \sqrt{|2\phi_F|} \right)$$

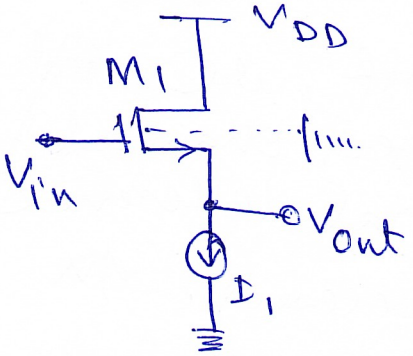
where V_{THO} denotes the threshold voltage with $V_{SB} = 0$, and $\gamma = \sqrt{2q\epsilon_{Si} N_{sub}}/C_{ox}$ denotes the body effect coefficient. V_{SB} is the source bulk potential difference.

Typical value of $\gamma = 0.4\sqrt{V}$ and $\phi_F = 0.4 V$.

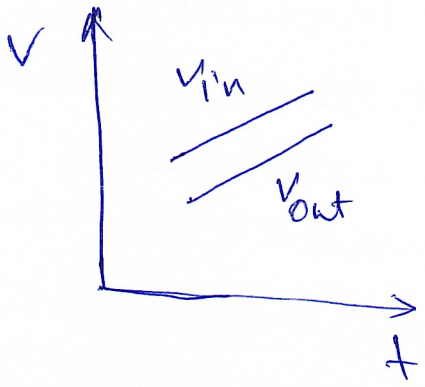
For body effect to manifest itself, the bulk potential, V_{sub}, need not change: if the source voltage varies with respect to V_{sub}, the same phenomenon occurs. For example, consider the circuit in Fig below, first ignoring body effect. We note that as V_{in} varies, V_{out} closely follows the input because the drain current remains equal to I_1. In fact we can write

$$I_1 = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{in} - V_{out} - V_{TH})^2$$

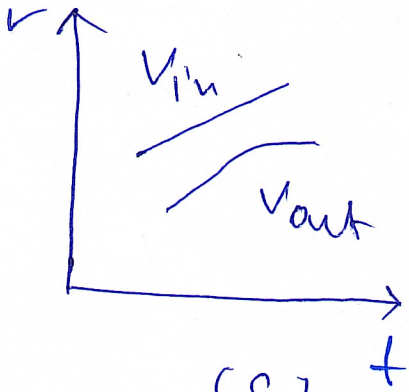
Concluding that $V_{in} - V_{out}$ is constant if I_1 is constant (Fig. b).



(a)



(b)



(c)

Now suppose the substrate is tied to ground and body effect is significant. Then, as V_{in} and hence V_{out} become more positive, the potential difference betn the source and the bulk increases, raising the value of V_{TH}. Therefore $V_{in} - V_{out}$ must increase so as to maintain I_D constant. Body effect is usually undesirable. The change in the threshold voltage, e.g. gain in Fig a, often complicates the design of analog circuits. Device technologists balance N_{sub} and C_{ox} to obtain a reasonable value for V_{TH}.

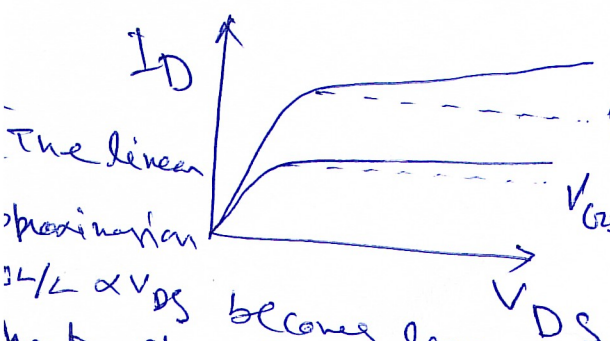
Channel-Length Modulation

In the analysis of channel pinch-off, we noted that the actual length of the inverted channel gradually decreases as the potential difference betⁿ the gate and the drain increases. In fact L' is a function of v_{DS}. This effect is called "channel-length modulation."

Writing $L' = L - \Delta L$, i.e. $1/L' \approx (1 + \Delta L/L)/L$, and assuming a first-order relationship betⁿ $\Delta L/L$ and v_{DS} such as $\Delta L/L = \lambda v_{DS}$, we have in saturation,

$$I_D \approx \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (v_{GS} - V_{TH})^2 (1 + \lambda v_{DS})$$

where λ is the channel-length modulation coefficient. This phenomenon results in a non-zero slope in the I_D/v_{DS} characteristics and hence a nonideal current source both D and S in saturation. The parameter λ represents the relative variation in length for a given increment in v_{DS}. Thus, for longer channels, λ is smaller.



The linear approximation $\Delta L/L \propto v_{DS}$ becomes less accurate in short-channel transistors.

* With channel length modulation, some of the expression derived for g_m must be modified.

$$g_m = \mu_n C_{ox} \frac{W}{L} (v_{GS} - V_{TH})(1 + \lambda v_{DS})$$

$$= \sqrt{\frac{2\mu_n C_{ox} (W/L) I_D}{1 + \lambda v_{DS}}}$$

at $g_m = \frac{2I_D}{v_{GS} - V_{TH}}$ remains unchanged!

Subthreshold Conduction:-

On our analysis of the MOSFET, we have assum that the device turns off abruptly as V_{GS} drops below V_{TH}. In reality, for $V_{GS} \approx V_{TH}$, a "weak" inversion layer still exists and some current flo from D to S. Even for $V_{GS} < V_{TH}$, I_D is finite but it exhibits an exponential dependence on V_{GS}. Called "Subthreshold Conduction" this effe can be formulated for V_{DS} greater than roughly 200 mV as

$$I_D = I_0 \exp \frac{V_{GS}}{\xi V_T} \quad \ldots\ldots$$

where $\xi > 1$ is a non-ideality factor and $V_T = \frac{KT}{q}$.

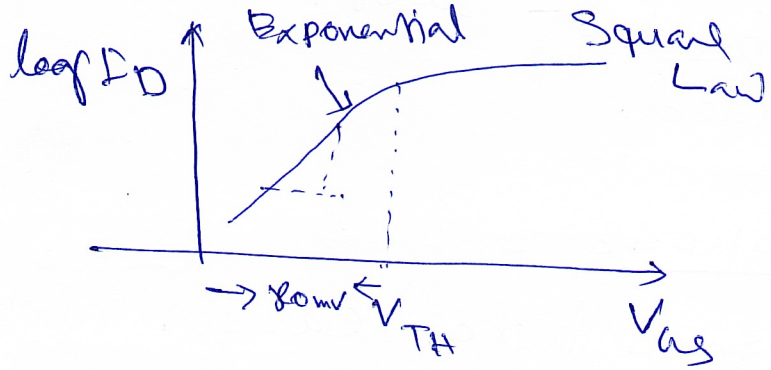


Fig. MOS Subthreshold Characteristics.

We now re-examine for the transconductance of a MOS device operating in the subthreshold region. Is it possible to achieve an arbitarily high transconductance by increasing W to while maintaining I_D Constant? Is it possible to obtain a higher

transconductance than that of a bipolar transistor (I_C/V_T) biased at the same current, we know $I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH})^2$. However, if W increases while I_D remains constant, then $V_{GS} \to V_{TH}$ and the device enters the subthreshold region. As a result, the transconductance is calculated to be $g_m = I_D/\xi V_T$, revealing that MOSFET are inferior to bipolar transistors in this respect.

The exponential dependence of I_D upon V_{GS} in subthreshold operation may suggest the use of mos devices in this regime so as to achieve a higher gain. However, since such conditions are met by only a large device width or low drain current, the speed of subthreshold circuits is severely limited.

Velocity Saturation:

At high electric fields, carrier mobility degrades, eventually leading to a constant velocity. Owing to the very short channels (e.g. 0.1 μm), modern mos devices experience velocity saturation even with drain-source voltages as low as 1V. As a result, the I-V characteristics no longer follow the square-law behavior.

We have $I_D = v_{sat} Q_1 = v_{sat} \cdot W C_{ox}(V_{GS}-V_{TH})$

I_D now exhibits a linear dependence on $V_{GS}-V_{TH}$ and no dependence on L. We also write $g_m = \frac{\partial I_D}{\partial V} =$